The empirical recurrence for OEIS A238177: recovering a conjecture that is not written down

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Abstract

OEIS A238177 records "Empirical recurrence of order 84", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 859$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 84, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 84 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A238177 is "Number of (n+1)X(5+1) 0..2 arrays with the maximum plus the minimum of every 2X2 subblock equal.". It has offset 1 and begins

$$79641, 12951743, 1917124097, 296231759327, 45437916131409, 6990206222515407, 1074790095990669977, 165289$$

The entry states:

> Empirical recurrence of order 84 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 10:48:25, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 859$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 859.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1718$ exact terms of the model returns order 84 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{84} c_i \, a(n-i),$$

c_{1}	330	c_{22}	1776031217711470948836889378324	c_{43}	49
c_{2}	−35999	c_{23}	48311529156467726464034844435546	c_{44}	98
c_{3}	1179501	c_{24}	−344421381758520413120079624743306	c_{45}	−5
c_{4}	62688612	c_{25}	−1626465713024807053007774643520900	c_{46}	−3
c_{5}	−6051747491	c_{26}	25919942206044252740086656330035538	c_{47}	41
c_{6}	114098359394	c_{27}	−10807712914777954075502827945601996	c_{48}	−48
c_{7}	4965981030303	c_{28}	−1155539004509093442849311333310525560	c_{49}	−19
c_{8}	−264592223808097	c_{29}	4113809405263010813199480960606871258	c_{50}	476
c_{9}	2234621301255526	c_{30}	31025958439229240800520183366355338196	c_{51}	438
c_{10}	128391326133386141	c_{31}	−226330108361094525799960336029904522856	c_{52}	−254
c_{11}	−3559690806488339940	c_{32}	−353315890909595890583147442741334533069	c_{53}	710
c_{12}	3457015184907797404	c_{33}	7175093256127139814222728776624483662130	c_{54}	8505
c_{13}	1171536676377756360401	c_{34}	−8049609691230525773977807058783376607779	c_{55}	−977
c_{14}	−15357015103378704319350	c_{35}	−146264830134642522792803357113397832799499	c_{56}	−166
c_{15}	−114237697977837776237474	c_{36}	478576251375357574069307591394325871641924	c_{57}	3883
c_{16}	4227635315073498122689046	c_{37}	1797242393425021674379301853748620406999129	c_{58}	949
c_{17}	−16420048903214560667341156	c_{38}	−11682316824214848509239034219946049735998070	c_{59}	−884
c_{18}	−512233756046839916751420166	c_{39}	−6667618289570064025466910649842963352170513	c_{60}	4279
c_{19}	5748111599748089669976098102	c_{40}	177633276646469699087126135696955794818216267	c_{61}	1199
c_{20}	22541957351611303463568869688	c_{41}	−210088488443313028248007317216550316562437122	c_{62}	−1375
c_{21}	−714693968298085681094840455798	c_{42}	−1751927748316401014409590761388969398781923695	c_{63}	−775

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 84, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $84 + 4$ terms removed returns order 84 as well, so $\deg q_{\min} = 84 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 11 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $84+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 84 that this sequence satisfies — and that family turns out to have exactly one member.

References

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.